

COM748 Masters Research Project

Project Proposal Form

Student Name	Student Number	Supervisor Name

Project Title

Explainable Transformer Based Framework for Early Detection of Mental Health Disorders from Social Media Text

Problem Statement (100 words)

Mental health disorders such as depression, anxiety, and stress are becoming increasingly prevalent worldwide and represent a significant public health challenge. Early diagnosis is often difficult because traditional assessment methods depend on clinical interviews, self reported questionnaires, and observations by healthcare professionals. Social media platforms such as Reddit provide a rich source of textual data where users frequently express emotions, experiences, and psychological concerns. Advances in Artificial Intelligence and Natural Language Processing have created opportunities to analyse such data for early identification of mental health risks. This project aims to develop an explainable transformer based framework capable of detecting mental health disorders from social media text while providing transparent and interpretable predictions. The proposed system seeks to support healthcare professionals and researchers by enabling scalable, automated, and trustworthy mental health screening.

Background Research (500 words)

Mental health disorders affect millions of people globally and continue to place considerable pressure on healthcare systems. Depression, anxiety, and stress are among the most common conditions, and delayed diagnosis often leads to worsening symptoms and reduced quality of life. Consequently, researchers have increasingly explored Artificial Intelligence based approaches to support early identification and intervention.

Recent studies have demonstrated that Natural Language Processing techniques can identify indicators of mental health disorders from social media communications. Patnaik et al. [1] compared CNN, XGBoost, BRDT, and Naive Bayes models for analysing social media content and demonstrated promising performance in detecting depression related patterns. Their work highlights the potential of machine learning methods for automated mental health assessment.

Aragon et al. [2] investigated emotional patterns in social media posts to detect depression and anorexia. Their findings revealed that emotional and linguistic characteristics are valuable predictors of mental health conditions. Similarly, Herdiansyah et al. [3] applied sentiment analysis techniques to identify mental health issues from social media data and demonstrated that textual information can provide important insights into psychological wellbeing.

The rapid development of transformer based architectures has significantly improved text classification performance. Unlike traditional machine learning models that rely heavily on

handcrafted features, transformer models learn contextual relationships between words and capture semantic information more effectively. Recent surveys indicate that transformer based models such as BERT consistently outperform traditional machine learning and deep learning approaches in Natural Language Processing tasks [5].

Despite these advances, many Artificial Intelligence systems remain difficult to interpret. In healthcare related applications, transparency is essential because users and practitioners must understand why predictions are made. Explainable Artificial Intelligence techniques such as SHAP and LIME have emerged as effective methods for improving trust and interpretability in machine learning systems. Olawade et al. [4] highlighted the growing importance of explainable and trustworthy Artificial Intelligence solutions within healthcare environments.

Several publicly available datasets support research in this area. The Social Media Mental Health Dataset (SMHD) [6] contains user generated content associated with multiple mental health conditions and is widely used for mental health classification research. The Depression Reddit Cleaned Dataset [7] provides curated Reddit posts associated with depression related discussions, while the Sentiment Analysis for Mental Health Dataset [8] contains labelled social media content representing various emotional and psychological states. These datasets provide a valuable foundation for developing and evaluating robust mental health detection models.

While previous studies have focused primarily on binary classification or model performance, fewer studies have investigated explainable transformer based frameworks capable of providing transparent predictions for multiple mental health conditions. This project aims to address this gap through the development of an explainable transformer based system for early detection of mental health disorders from social media text.

Project Aim and Objectives (300 words)

Aim

To design, develop, and evaluate an explainable transformer based framework for early detection of mental health disorders from social media text and compare its performance against traditional machine learning and deep learning approaches.

Objectives

1. Conduct a comprehensive literature review on Artificial Intelligence, Natural Language Processing, transformer models, and Explainable Artificial Intelligence techniques for mental health detection.
2. Collect and preprocess mental health related social media datasets from Reddit and publicly available repositories.
3. Implement baseline machine learning models including Logistic Regression and XGBoost.
4. Develop deep learning models including Convolutional Neural Networks (CNN) and Bidirectional Long Short Term Memory (BiLSTM) networks.

5. Implement a transformer based model using Bidirectional Encoder Representations from Transformers (BERT).
6. Integrate SHAP and LIME techniques to provide transparent and interpretable predictions.
7. Evaluate and compare all models using Accuracy, Precision, Recall, F1 Score, ROC AUC, and Confusion Matrix metrics.
8. Develop a web based application that allows users to submit text and receive mental health predictions along with explanation visualisations.
9. Analyse the effectiveness of explainability methods in improving trust and transparency within mental health prediction systems.

Methodology/approach (400 words)

The proposed research will follow an experimental methodology involving data collection, preprocessing, model development, evaluation, and deployment.

Data Collection

Mental health related datasets will be collected from publicly available repositories. The primary datasets include the Social Media Mental Health Dataset (SMHD) [6], Depression Reddit Cleaned Dataset [7], and Sentiment Analysis for Mental Health Dataset [8]. These datasets contain labelled social media posts associated with depression, anxiety, stress, and other mental health conditions.

Data Preprocessing

The collected textual data will undergo several preprocessing steps to improve quality and consistency. These steps will include lowercasing, tokenisation, stop word removal, lemmatisation, emoji removal, URL removal, punctuation cleaning, and text normalisation. The objective is to reduce noise and improve model learning performance.

Feature Engineering

Two feature extraction approaches will be investigated. Traditional machine learning models will utilise TF IDF representations, while transformer models will generate contextual embeddings using BERT. This comparison will provide insights into the effectiveness of contextual language representations.

Model Development

Several models will be developed and compared:

- Logistic Regression

- XGBoost
- CNN
- BiLSTM
- BERT

Traditional machine learning and deep learning models will serve as baseline approaches, while BERT will represent the primary transformer based architecture.

Explainable Artificial Intelligence

SHAP and LIME techniques will be integrated into the framework to explain individual predictions and identify the most influential words and phrases contributing to classification outcomes. This component will enhance transparency and trustworthiness.

Evaluation

The models will be evaluated using standard classification metrics including Accuracy, Precision, Recall, F1 Score, ROC AUC Score, and Confusion Matrix analysis. Comparative analysis will determine the strengths and limitations of each approach.

System Development

A web based application will be developed using Streamlit or React with a FastAPI backend. Users will be able to input text and receive mental health predictions accompanied by explanation visualisations generated through SHAP and LIME.

Software and Hardware

The implementation will utilise Python together with TensorFlow, PyTorch, Scikit Learn, HuggingFace Transformers, NLTK, spaCy, SHAP, and LIME libraries. GPU acceleration through Google Colab or cloud based resources may be employed to support transformer model training.

Legal/ Social/ Ethical Issues (200 words)

The project will adhere to the ACM Code of Ethics and the BCS Code of Conduct throughout all stages of development and evaluation. Since the research involves mental health related data, particular attention will be given to privacy, transparency, fairness, and responsible Artificial Intelligence practices.

Only publicly available and anonymised datasets will be used. No attempts will be made to identify individual users or extract personal information from social media platforms. All collected data

will be stored securely and used exclusively for academic research purposes.

Mental health prediction systems may produce incorrect classifications. False positives and false negatives could potentially lead to misinterpretation of user wellbeing. Therefore, the developed framework will not be presented as a clinical diagnostic tool but rather as a decision support system intended to assist researchers and healthcare professionals.

The incorporation of Explainable Artificial Intelligence techniques aims to improve transparency and allow users to understand the reasoning behind predictions. Any ethical approval requirements identified by the university and project supervisor will be completed before commencing the research.

Project & Risk Management (300 words)

Effective project management is essential to ensure successful completion of the project within the allocated timeframe.

Health and Safety Risks

The project primarily involves software development and data analysis activities using standard computing equipment. The associated risks are minimal and mainly relate to prolonged computer usage. Appropriate workstation ergonomics and regular breaks will be adopted to minimise discomfort and fatigue.

Management Risks

A significant risk is insufficient time for implementation, testing, and evaluation. Delays during model development may affect later stages of the project. To mitigate this risk, project activities will be divided into clearly defined milestones with regular progress reviews conducted with the project supervisor.

Another management risk involves scope expansion. The project will maintain focus on predefined objectives and avoid unnecessary features that could impact completion.

Technical Risks

Transformer models require considerable computational resources and training time. Limited access to GPU resources may affect experimentation. This risk will be mitigated through the use of cloud based platforms such as Google Colab.

Dataset quality and class imbalance may also affect model performance. Multiple datasets will be evaluated to ensure adequate data availability and diversity. Data balancing and augmentation techniques will be considered if necessary.

There is also a risk that model performance may not achieve expected results. To mitigate this issue, multiple machine learning, deep learning, and transformer models will be implemented and

compared. Continuous evaluation and validation will be performed throughout development.

References in IEEE format (Approx 5 References)

- [1] S. Patnaik, K. Sharma, I. Mohanty, and H. S. Pradhan, "AI Powered Sentiment Analysis for Early Detection of Mental Health Disorders in Social Media Communications," in 2025 Tenth International Conference on Science Technology Engineering and Mathematics (ICONSTEM), IEEE, 2025, pp. 1 to 6.
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- [3] H. Herdiansyah, R. Roestam, R. Kuhon, and A. S. Santoso, "Their Post Tell the Truth: Detecting Social Media Users Mental Health Issues with Sentiment Analysis," Procedia Computer Science, vol. 216, pp. 691 to 697, 2023.
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- [5] U. Singh, K. Abhishek, and H. K. Azad, "A Survey of Cutting Edge Multimodal Sentiment Analysis," ACM Computing Surveys, vol. 56, no. 9, pp. 1 to 38, 2024.
- [6] A. Cohan, S. Desmet, A. Yates, L. Soldaini, A. Cohan, and N. Goharian, "SMHD: A Large Scale Resource for Exploring Online Language Usage for Multiple Mental Health Conditions," in Proceedings of the 27th International Conference on Computational Linguistics, Santa Fe, NM, USA, 2018, pp. 1485–1497.
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- [8] S. Sarkar, "Sentiment Analysis for Mental Health," Kaggle, 2022. [Online]. Available: <https://www.kaggle.com/datasets/suchintikasarkar/sentiment-analysis-for-mental-health>. Accessed: Jun. 2, 2026.